

Comprehensive System Feature Specification

Project Name: Secure AI-Powered Note Management System **Architecture:** Next.js (Frontend) | Spring Boot (Backend) | PostgreSQL (Database)

1. User Interface (UI) & Dashboard Layout

The UI is designed for high productivity with a modern, responsive interface.

A. Home Dashboard

- **Latest Groups Section:** Displays a horizontal scroll or grid of the most recently accessed Groups (formerly Folders) for quick navigation.
- **Latest Documents Section:** A dedicated area showing the user's most recently edited or created notes for immediate access.
- **Analytics Overview:** (Optional) Brief stats on total notes, shared notes, and pending reminders.

B. Navigation Sidebar

- **Recent:** A chronological list of all documents accessed recently.
- **Shared with Me:** A specialized section for documents and groups shared by other users of the application.
- **Favorites:** Quick access to notes and groups marked with a star/favorite icon.
- **Bin (Recycle Bin):** Access to "Soft Deleted" items pending permanent removal.
- **Activities:** A dedicated log view for tracking all system actions.

2. Note & Group Management (Organization)

The system moves away from traditional folder terminology to a "**Group**" based hierarchy.

- **Group Concept:** Users organize notes into "Groups" (Folders) which support nested sub-groups.
- **Comprehensive Item Actions:** Every document and group supports the following 11 actions:
 1. **Lock:** Apply a secondary password for sensitive content.
 2. **Share:** Manage collaborator access and permissions.
 3. **Download:** Export as PDF, Markdown, or TXT.
 4. **Move:** Relocate an item between different Groups.
 5. **Remove:** Send item to the Bin (Soft Delete).
 6. **Copy:** Create a copy in a different location.
 7. **Duplicate:** Immediate "Copy-Paste" in the same location.

8. **Make Version:** Manually trigger a Git-style snapshot.
9. **Archive:** Hide from the active dashboard while preserving data.
10. **Add Favorite:** Pin to the Favorites section for priority access.
11. **Reminder:** Set a specific date/time for system notifications.

3. AI-Powered Smart Features

Artificial Intelligence is integrated to reduce manual organization effort.

- **Smart Tagging:** Automatically suggests 3-5 relevant tags based on the note's text.
- **Category Prediction:** AI predicts the best category (e.g., Work, Study, Personal) for the note.
- **AI Summarization:** Generates concise summaries for long-form notes to help users skim content quickly.
- **Manual AI Trigger:** Users can choose when to invoke AI assistance to maintain control.

4. Git-Style Manual Version Control

This feature allows for clear milestone tracking and prevents database clutter.

- **Manual Snapshots:** Users create versions manually with custom labels (e.g., "SRS v1", "Final Review").
- **Version History Sidebar:** A dedicated panel showing a timeline of all saved versions.
- **Preview Mode:** Users can safely view old content without overwriting their current live draft.
- **One-Click Restore:** Revert the entire document to any previous snapshot instantly.

5. Security & Privacy Framework

Enhanced protection through a dual-security layer.

- **Encrypted Storage:** Note content is encrypted at the database level to ensure privacy.
- **Per-Item Passwords:** Users can secure any specific Note or Group with a unique password.
- **Visibility Control:** Toggle visibility between Private (owner only) and Shared.
- **JWT Authentication:** Secure user registration, login, and profile management.

6. Collaboration & Sharing (ACL)

Built for professional teamwork and transparency.

- **Role-Based Access Control:**

- **Owner:** Full control over the note and its permissions.
 - **Editor:** Can modify content and metadata.
 - **Viewer:** Read-only access.
 - **Viewer Tracking:** Owners can see a list of users who have viewed their shared documents.
 - **Friends System:** A built-in directory for quick sharing with frequent collaborators.
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7. **Smart Bin System (Data Recovery)**

- **Soft Delete:** Documents are not immediately erased; they move to a "Pending Deleted" state.
 - **1-Month Retention:** Items stay in the Bin for 30 days before permanent deletion.
 - **Restoration:** Users can restore any item in the Bin to its original location with full history intact.
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8. **Activity Tracking & Notifications**

- **Global Activity Log:** Tracks every major action in the app, including edits, sharing events, and version restores.
- **Collaboration Updates:** Notifications for when a shared note is edited or a new collaborator is added.